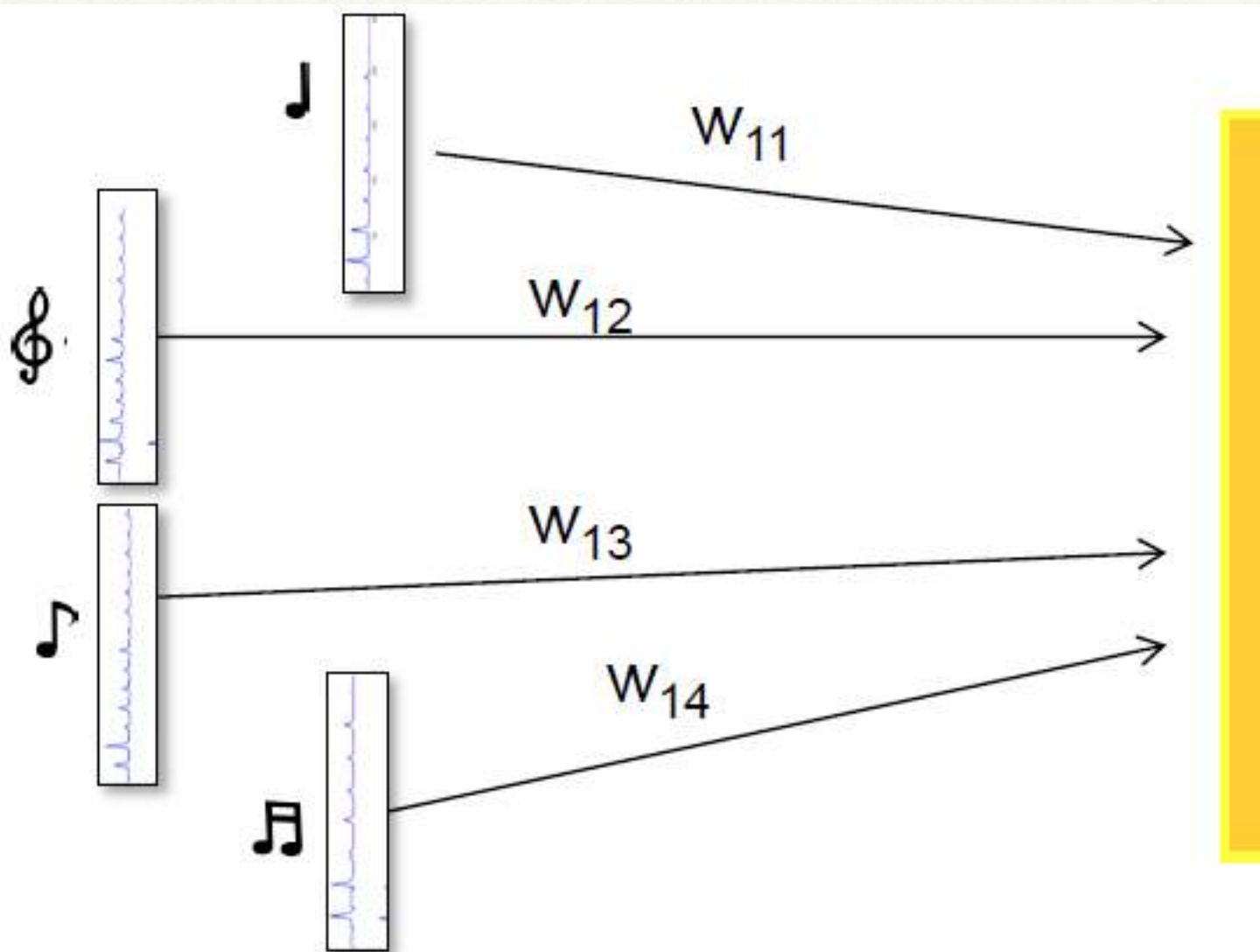


E9 205 Machine Learning for Signal Processing

Non-negative Matrix Factorization

10-08-2016

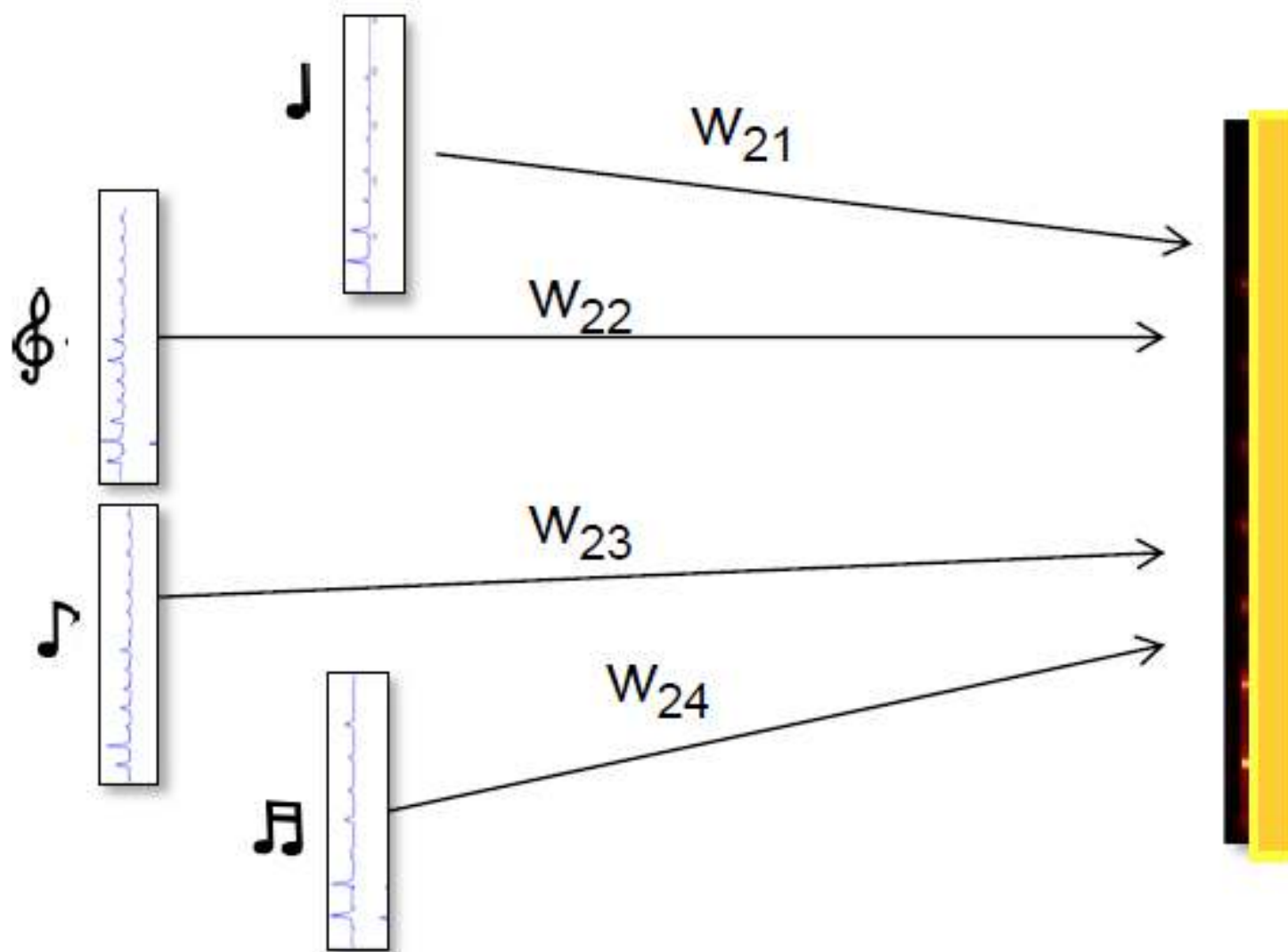
Audio Composition



- ❖ Audio signal generated by a number of individual components (example - different instruments).

<http://mlsp.cs.cmu.edu/courses/fall2014/lectures/slides/class10.nmf.pdf>

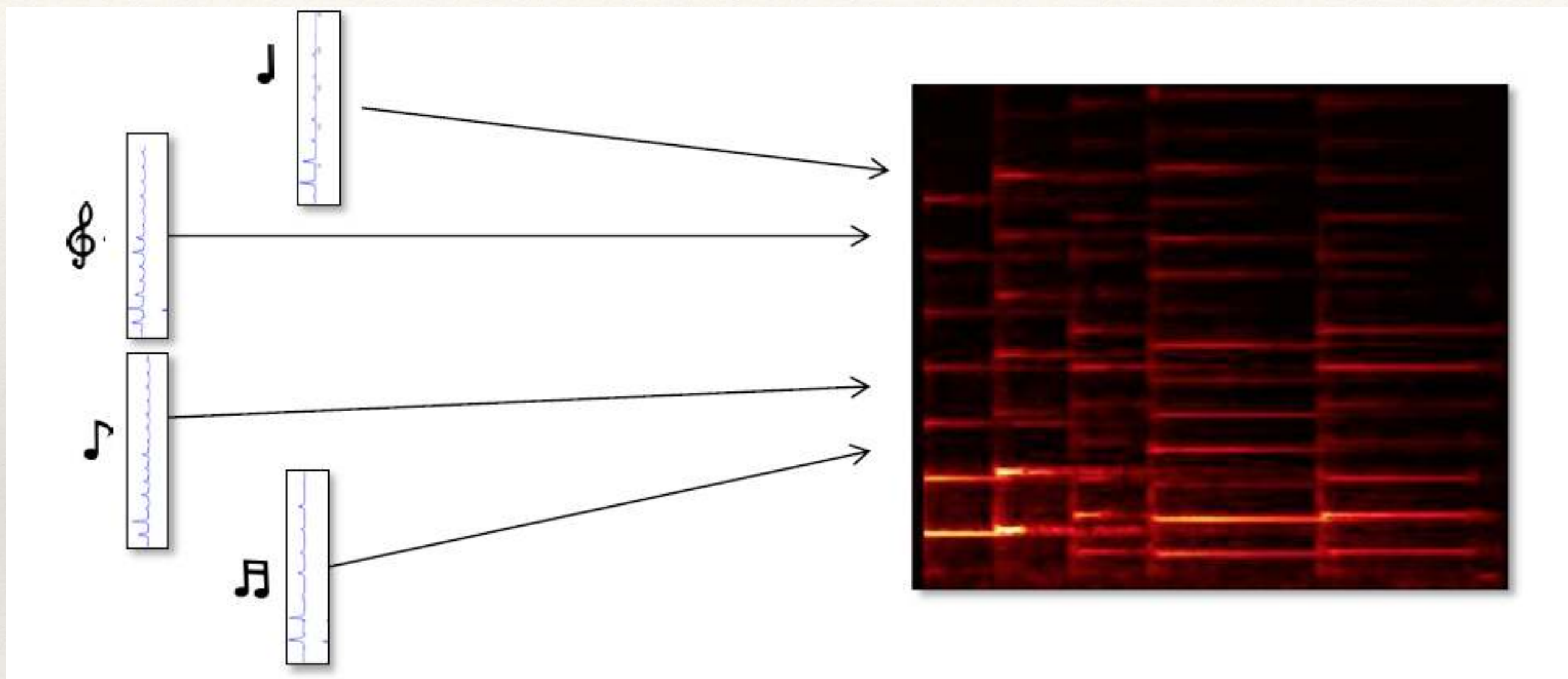
Audio Composition



- ❖ Audio signal generated by a number of individual components (example - different instruments).

<http://mlsp.cs.cmu.edu/courses/fall2014/lectures/slides/class10.nmf.pdf>

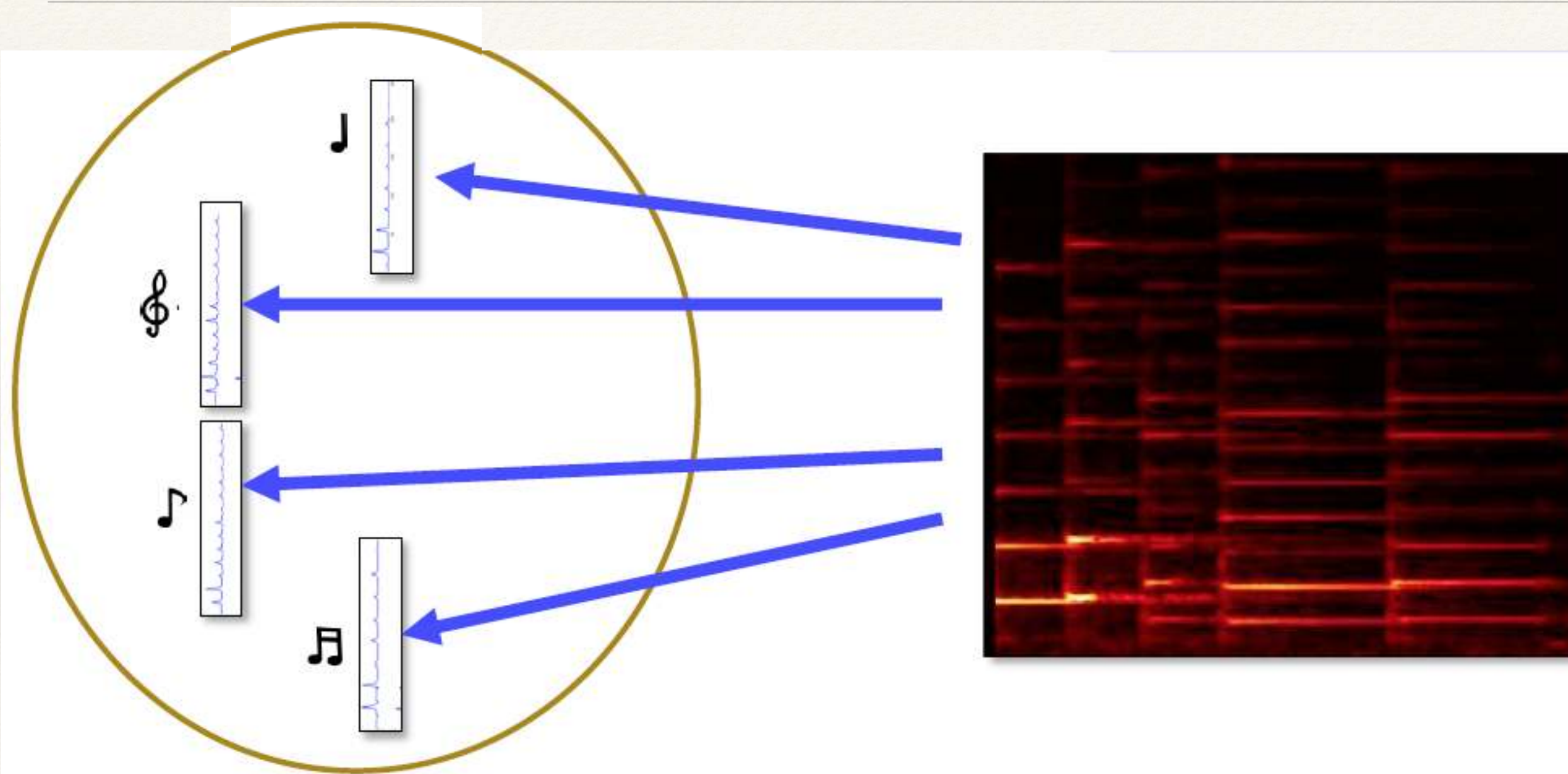
Audio Composition



- ❖ Audio signal generated by a number of individual components (example - different instruments).

<http://mlsp.cs.cmu.edu/courses/fall2014/lectures/slides/class10.nmf.pdf>

Audio Decomposition



- ❖ Decomposing an audio signal back to its fundamental components.

Required Properties for Decomposition

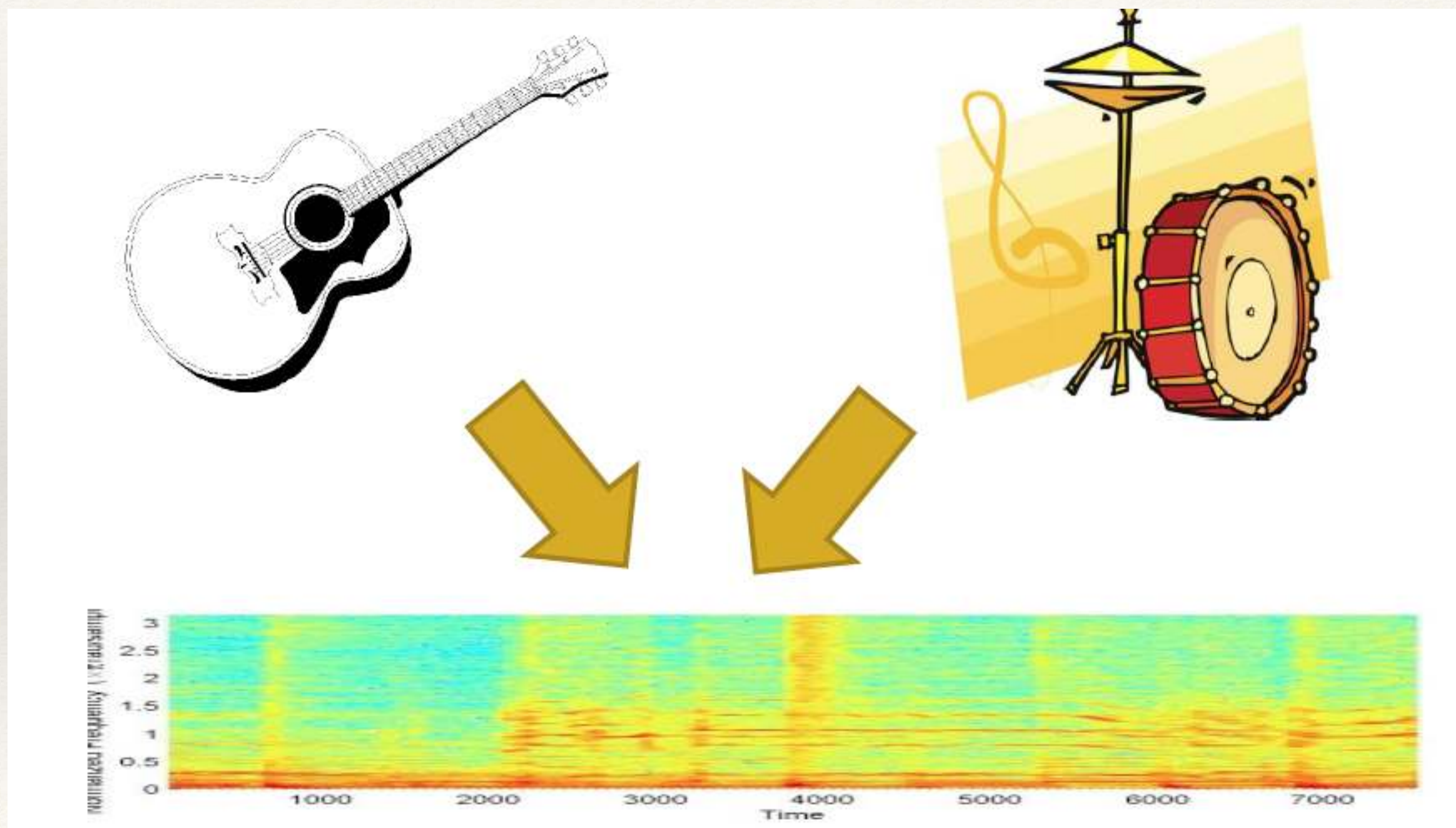
- ❖ Given input is sequence of power spectral vectors (positive) - power spectrogram $\mathbf{V}_{k \times n}$
- ❖ Decomposition into a sequence of component power spectral vectors (positive) - basis $\mathbf{b}_{k \times 1}^j$
- ❖ At each frame - non-negative combination of components are allowed - +ve weights $\mathbf{w}_{j \times 1}$

$$\mathbf{V} = \mathbf{B}\mathbf{W}$$

- ❖ Applying NMF to decompose the components

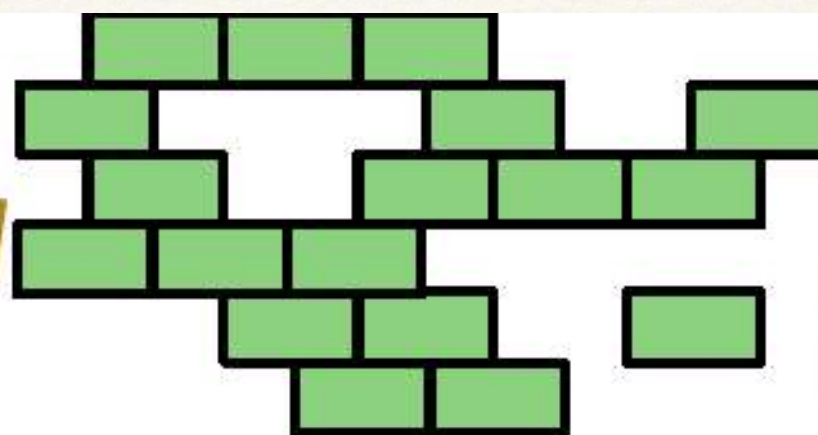
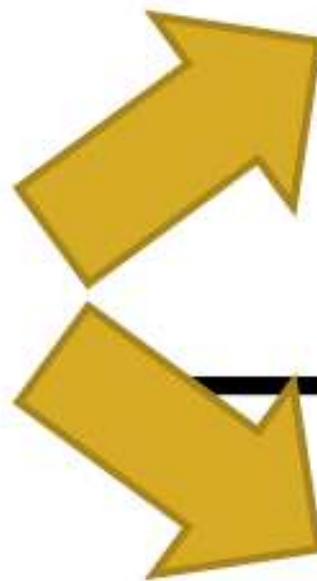
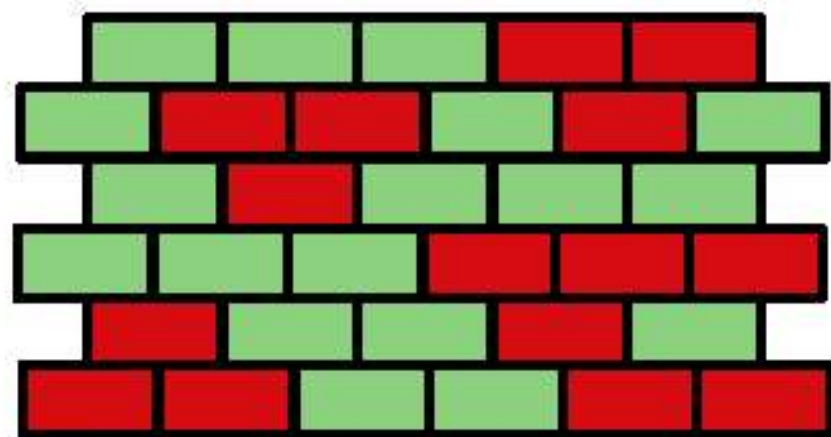
NMF Audio Example

Audio from two sources

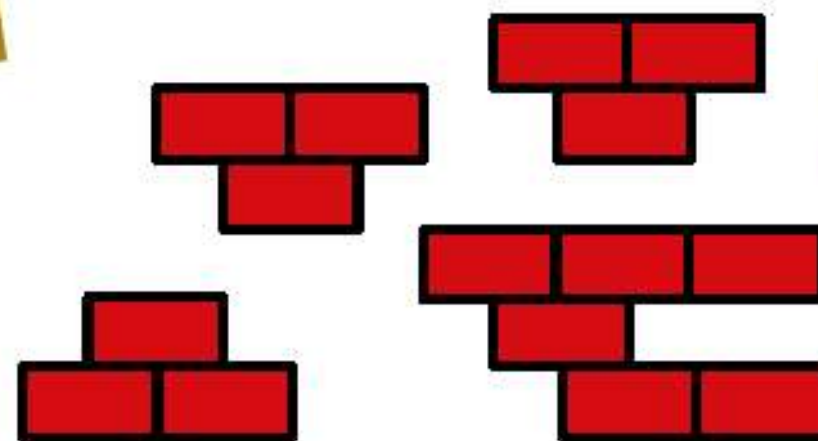


Audio from two sources

Composition



From green blocks

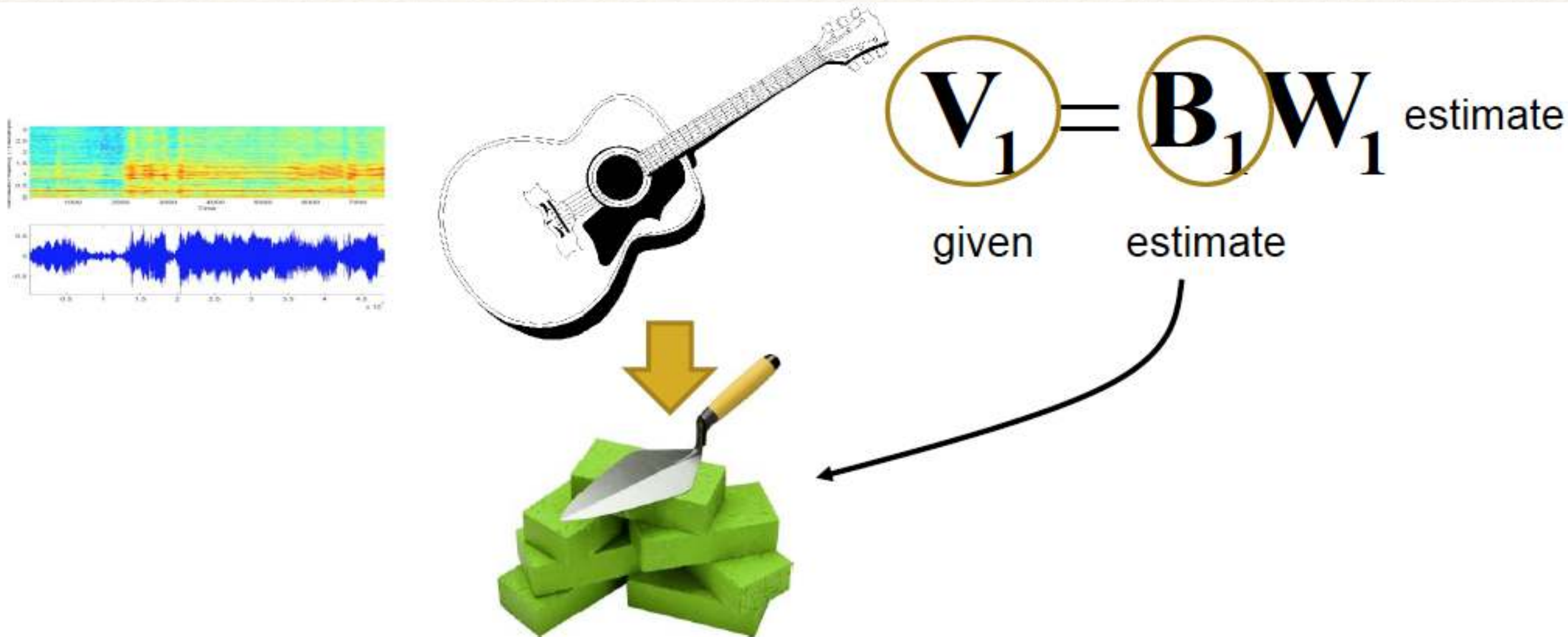


From red blocks

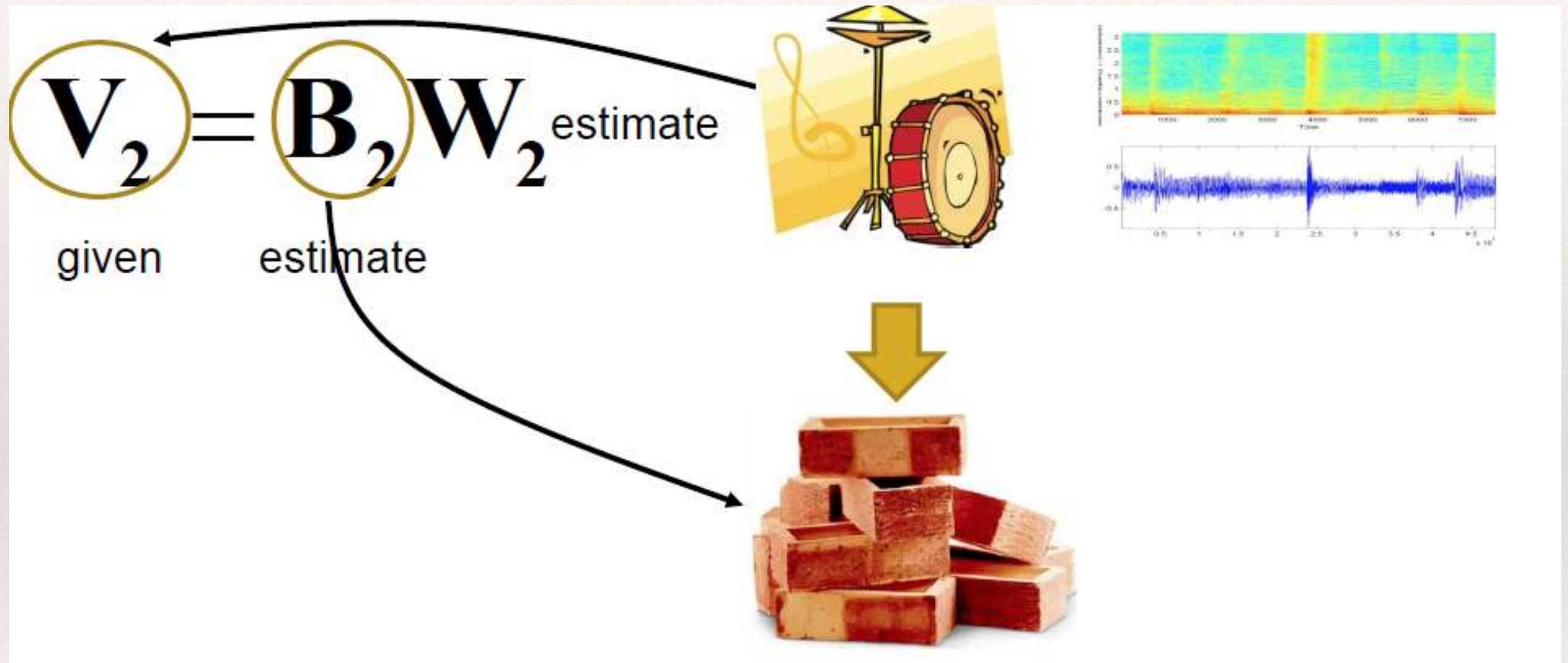


Building blocks

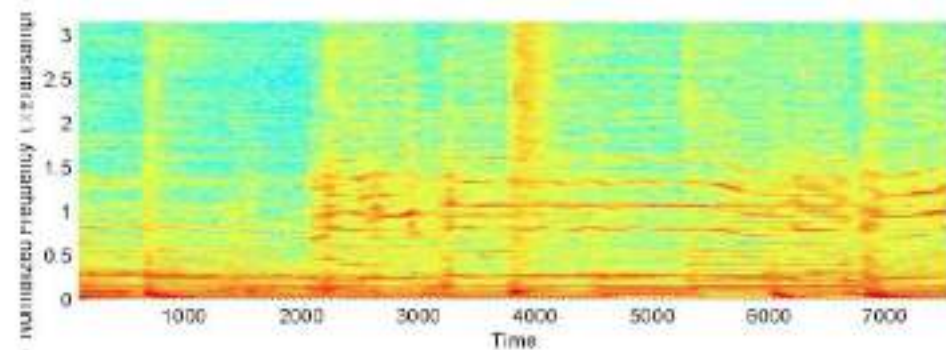
Basis Estimation from Source I



Basis Estimation from Source II



Finding the Mixing Weights



given



$$V = BW$$

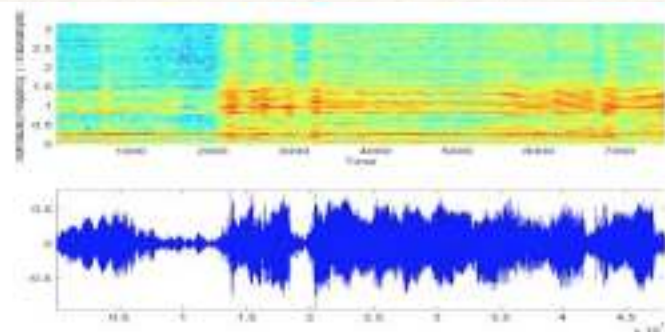
$$\begin{bmatrix} B_1 & B_2 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

given

estimate

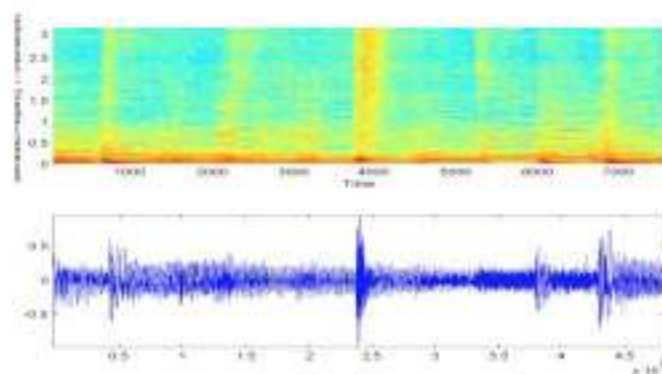


Finding the Mixing Weights - Known Sources



estimate

$$\mathbf{B}_1 \mathbf{W}_1$$



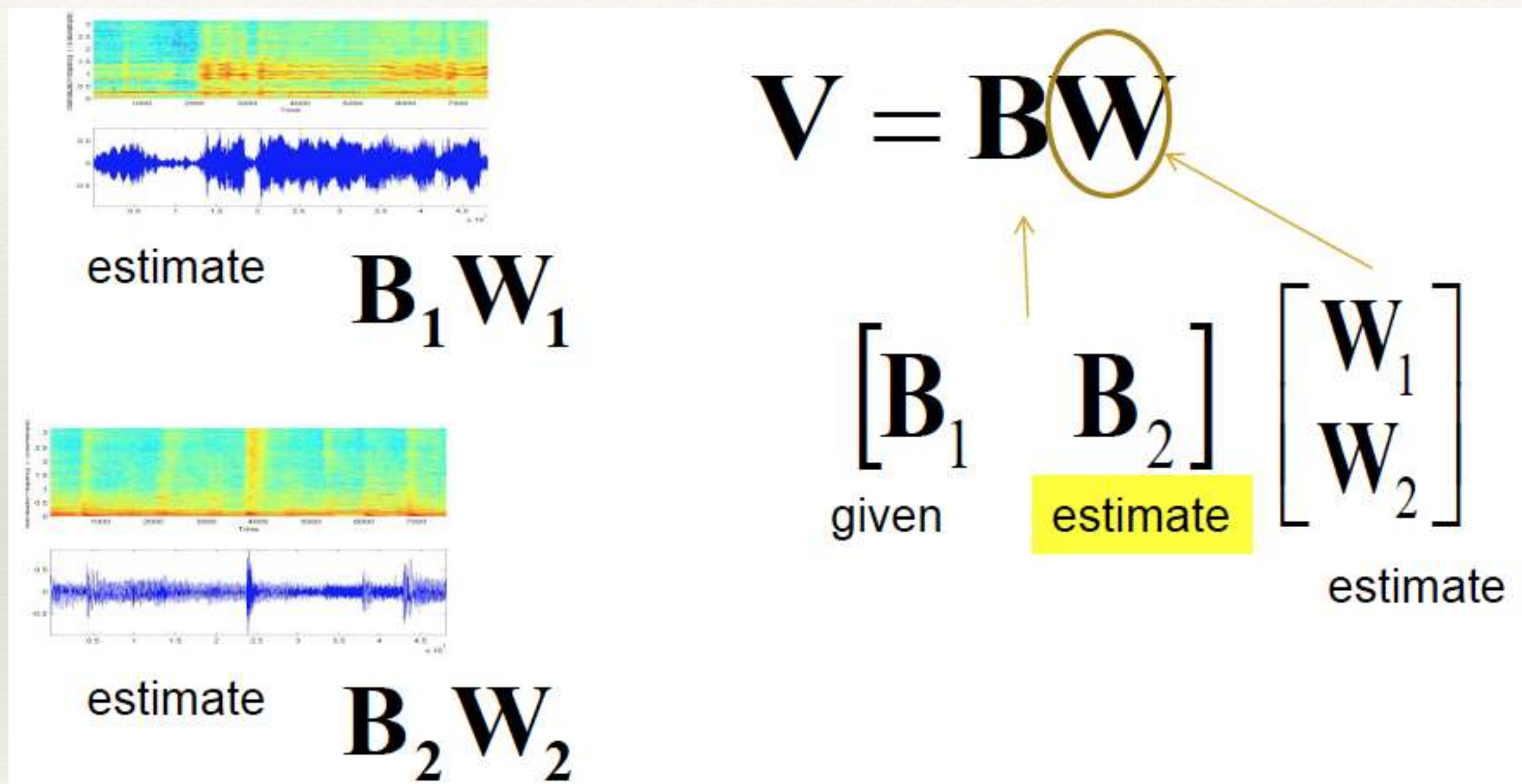
estimate

$$\mathbf{B}_2 \mathbf{W}_2$$

$$\mathbf{V} = \mathbf{B} \mathbf{W}$$

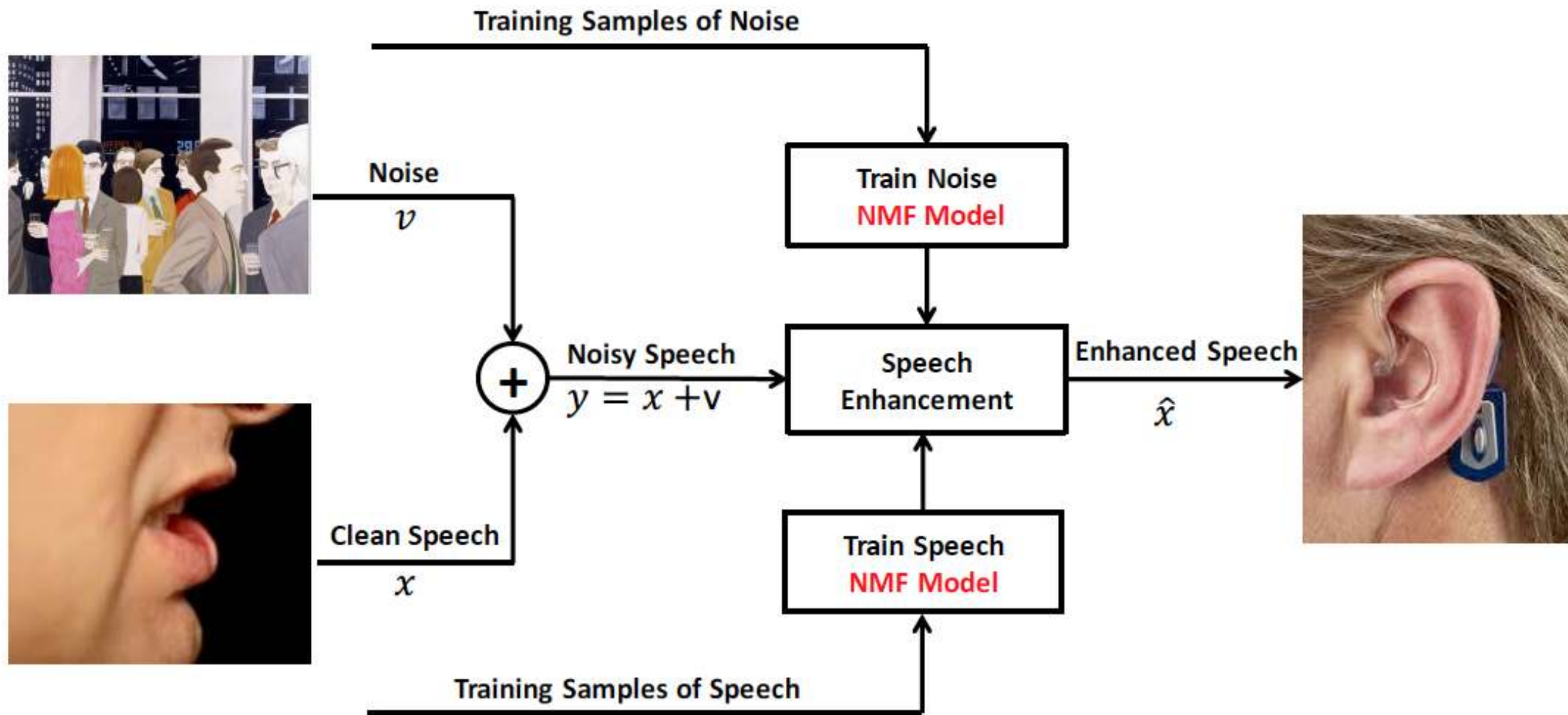
Diagram illustrating the matrix equation $\mathbf{V} = \mathbf{B} \mathbf{W}$. The matrix $\mathbf{B}$ is composed of two submatrices, $\mathbf{B}_1$ and $\mathbf{B}_2$, which are labeled "given". The vector $\mathbf{W}$ is composed of two subvectors, $\mathbf{W}_1$ and $\mathbf{W}_2$, which are labeled "estimate". A yellow circle highlights the product $\mathbf{B} \mathbf{W}$, with arrows pointing from the "given" and "estimate" labels to the corresponding matrices and vectors in the equation.

Finding the Mixing Weights - Unknown Source-II

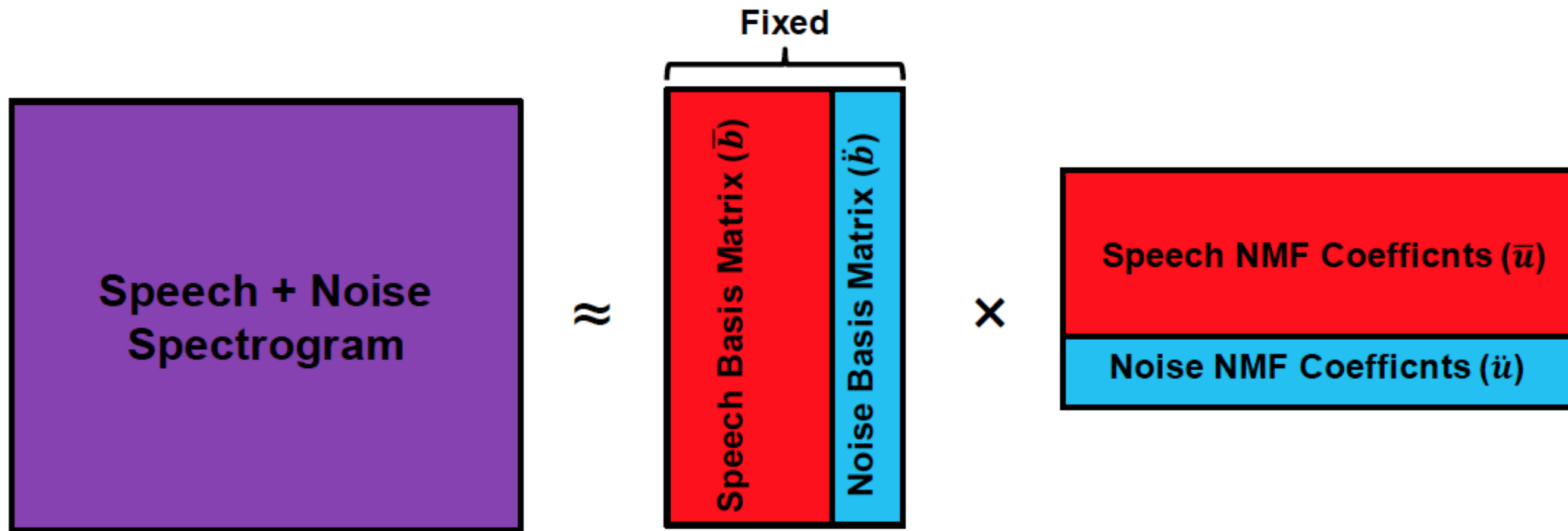


NMF Speech Denoting

Denoising using NMF



Denoising using NMF



Denoising using NMF

